



# Problem Solving With C - UE25CS151B

## Arrays and Functions

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# PROBLEM SOLVING WITH C

## Arrays and Functions

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1. Passing an Array to a function
2. Array as a Formal parameter and actual parameter
3. Pointer as a Formal parameter and Array as an Actual parameter
4. Usage of const keyword
5. Passing individual array element to a function

### Passing an Array to a function

- When array is passed as an argument to a function, arguments are copied to parameters of the function and parameters are always pointers.
- Array degenerates to a pointer at runtime.
- All the operations that are valid for pointer will be applicable for array too in the body of the function.
- Function call happens always at run time.

### Array as a formal parameter and Actual parameter

- **Array being formal parameter** - Indicated using empty brackets in the parameter list.

```
void myfun(int a[],int size);
```

- **Array being actual parameter** – Indicated using the name of the array  
Array a is declared as `int a[5];`  
Then myfun is called as `myfun(a,n);`
- Coding example to read and display the array elements

### Pointer as a formal parameter and Array as an Actual parameter

- **Pointer being formal parameter** - Indicated using empty brackets in the parameter list.  

```
void myfun(int *a,int size);
```
- **Array being actual parameter** – Indicated using the name of the array  
Array a is declared as `int a[5];`  
Then myfun is called as `myfun(a,n);`
- Coding example to read and display the array elements

### Usage of const keyword

- Array being the formal parameter allows a function to change the elements of the array passed in the function call.
- If a function should not modify the elements in the array, use the keyword **const**
- Example:  

```
void show_the_world(const int a[ ], int size);
```
- The **compiler will issue an error if there is a statement inside the body of the function to modify the array elements**
- Coding examples

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### Passing individual Array elements to a function

- Indexed variables can be arguments to functions
- Program contains these declarations: `int a[10]; int i; void myfunc(int n);`
- Variables `a[0]` through `a[9]` are of type `int`, making below calls is legal  
`myfunc(a[0]);`  
`myfunc(a[3]);`  
`myfunc(a[i]);`                      `// i is between 0 and 9`



## THANK YOU

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